

# Pinhole Precision Challenge

Build the sharpest camera image with no lens.

**Win condition:** Best image clarity and concept explanation after one aperture redesign.

## THE SCIENCE YOU ARE USING

**Light travels in straight lines.** A pinhole camera has no lens. Light from each point of a scene travels in a straight line through a tiny hole and lands on the screen, forming an upside-down image. Because the rays cross at the hole, the picture is flipped.

**The aperture tradeoff.** A smaller hole makes a sharper image but a dimmer one; a bigger hole is brighter but blurrier. You optimize the aperture for the best balance of sharpness and brightness, then redesign once.

## HOW TO BUILD AND RUN IT

- 01 Build the viewer.** Make a tube or box with a foil aperture at one end and a paper screen at the other.
- 02 Punch a small hole.** Pierce a clean, tiny hole in the foil with a pin. Smaller is sharper.
- 03 Find the image.** Aim at a bright scene and look for the dim, upside-down image on the screen.
- 04 Tune the aperture.** Try a slightly larger or smaller hole. Balance sharpness against brightness.
- 05 Explain the optics.** Describe why the image is flipped and why hole size changes the result.

**Safety:** No direct sun viewing. Pins handled at a controlled station. Allergy: None.

## MATERIALS

|                         |              |
|-------------------------|--------------|
| Cardstock               | shared       |
| Foil                    | for aperture |
| Tape                    | shared       |
| Pushpins or paper clips | shared       |
| White screen paper      | per team     |
| Light source            | shared       |

## YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): \_\_\_\_\_

The ONE variable we are testing: \_\_\_\_\_

Baseline result: \_\_\_\_\_

After redesign: \_\_\_\_\_

DEFEND YOUR DESIGN

What evidence made you change your design? \_\_\_\_\_

If you ran it again, what is the ONE thing you would change? \_\_\_\_\_

HOW YOU SCORE (100 POINTS)

| SCORING CRITERION   | POINTS |
|---------------------|--------|
| Image clarity       | 35     |
| Concept explanation | 25     |
| Build quality       | 20     |
| Redesign            | 10     |
| Speed               | 10     |
| Total               | 100    |